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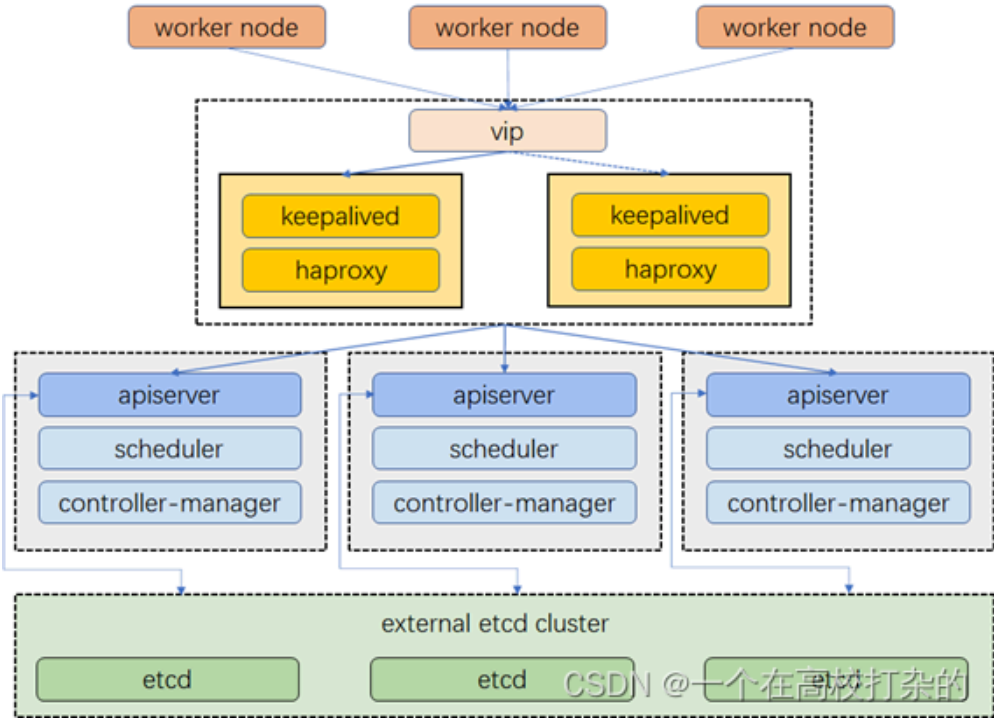
Article tags: #kubernetes #nginx #Container

QKCP LB Installation: High availability Kubernetes clusters ensure that applications run without service outages.

- Official deployment is provided by the customer, who is responsible for the customer data;
- LB resources need to be inspected and verified;
- The test environment can be built on virtual machines and provide storage resources.

Cluster architecture diagram:

The example cluster has three master nodes, three worker nodes, two nodes for load balancing, and a virtual IP address. The virtual IP address in this example may also be referred to as a “floating IP address”. This means that in the event of a node failure, the IP address can drift between nodes, resulting in high availability.



Note that in this example, Keepalived and HAproxy are installed on two separate nodes. Alternatively, configuring two specific nodes for load balancing separately (as shown in the figure above, you can add more such nodes on demand) will be more secure and flexible.

Configure Load Balancing

[Keepalived](#) Provides VRPP implementation and allows you to configure Linux machines to load balance and prevent single points of failure. [HAProxy](#) Provides reliable, high-performance load balancing that works perfectly with Keepalived.

Since Keepalived and HAproxy are installed on lb1 and lb2, if one of the nodes fails, the virtual IP address (i.e. floating IP address) will automatically be associated with the other node, so that the cluster can still operate normally, thus achieving high availability. If necessary, you can also add more nodes with Keepalived and HAproxy installed for this purpose.

Install Keepalived and HAproxy

ha01 and ha02 nodes install HAProxy and KeepAlived through apt:

apt install keepalived haproxy -y

The ha01 and ha02 nodes are configured with HAProxy (the HAProxy configuration of ha1 and ha2 nodes is the same):

vim/etc/haproxy/haproxy.cfg

HA proxy Configuration file: (More notes at source code Under /doc/intro.txt and /doc/configuration.txt)

root@prd-lb01:~# cat/etc/haproxy/haproxy.cfg

```
global
    maxconn 2000
    ulimit-n 16384
    log 127.0.0.1 local0 err
    stats timeout 30s
```

defaults

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0 0 0 0

mode http #working in layer 7, the most valuable part of Haproxy

in

option httplog

timeout connect 5000

timeout client 50000

timeout server 50000

timeout http-request 15s

timeout http-keep-alive 15s

frontend monitor-in

bind *:33305

mode http

option httplog

monitor-uri /monitor

frontend k8s-master

bind 0.0.0.0:16443

bind 127.0.0.1:16443

mode tcp #Works in layer 4, also default configuration, four layers of load

option tcplog

tcp-request inspect-delay 5s

default_backend qyk8s-master #Locate to the backend qyk8s-master

backend qyk8s-master

mode tcp #layer4must correspond to the front-end k8s-master

option tcplog

option tcp-check

balance roundrobin #load algorithm

default-server inter 10s downinter 5s rise 2 fall 2 slowstart 60s maxconn 250 maxqueue 256 weight 100

server host-master01 10.0.200.208:6443 check

server host-master02 10.0.200.209:6443 check

server host-master03 10.0.200.210:6443 check

KeepAlived Configuration File

The ha1 and ha2 nodes are configured with KeepAlived, but the configuration is different

Configuration of ha1 node:

mkdir/etc/keepalived

vim/etc/keepalived/keepalived.conf

```
root@prd-lb01:~# cat/etc/keepalived/keepalived.conf
! Configuration File for keepalived
global_defs {
    router_id LVS_DEVEL
}
vrrp_script chk_apiserver {
    script "/etc/keepalived/check_apiserver.sh"
    interval 5
    weight -5
    fall 2
    rise 1
```

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```
state MASTER #masterrole
interface ens3 #The network card name is ens3
# mcast_src_ip 10.0.200.217 #Send the address of the multicast package. If the default is not set, use the primary IP of the bound network card
virtual_router_id 151 #VRIDtag, route ID, viewable via #tcpdump vrrp
priority 101 #Priority, high priority election as master
advert_int 2
authentication {
  auth_type PASS #Authentication type
  auth_pass K8SHA_KA_AUTH #Authenticated password
}
virtual_ipaddress {
  10.0.200.221/24 #Set vip
}
track_script {
  chk_apiserver
}
}
```

Configuration of ha2 node:

```
root@prd-lb02:~# cat/etc/keepalived/keepalived.conf
! Configuration File for keepalived
global_defs {
  router_id LVS_DEVEL
}
vrrp_script chk_apiserver {
  script "/etc/keepalived/check_apiserver.sh"
  interval 5
  weight -5
  fall 2
  rise 1
}
vrrp_instance VI_1 {
  state BACKUP #Set to backup role
  interface ens3
  #mcast_src_ip 10.0.200.218
  virtual_router_id 151
  priority 100 #priority lower than master's 101
  advert_int 2
  authentication {
    auth_type PASS
    auth_pass K8SHA_KA_AUTH
  }
  virtual_ipaddress {
    10.0.200.221/24
  }
  track_script {
    chk_apiserver
  }
}
```

```
# track_script {
# chk_apiserver
# }
```

Configure the KeepAlived health check file:

cat/etc/keepalived/check_apiserver.sh

```
#!/bin/bash
err=0
for k in $(seq 1 3)
do
  check_code=$(pgrep haproxy)
  if [[ $check_code == "" ]]; then
```

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0 0 0 0 0

```
        continue
    else
        err=0
        break
    fi
done
if [[ $err != "0" ]]; then
    echo "systemctl stop keepalived"
    /usr/bin/systemctl stop keepalived
    exit 1
else
    exit 0
fi
```

Add script execution permissions:

```
chmod +x/etc/keepalived/check_apiserver.sh
```

Start haproxy and keepalived

```
systemctl restart haproxy
```

```
systemctl restart keepalived
```

```
systemctl enable haproxy && systemctl enable keepalived
```

HAProxy and KeepAlived Test

Important: After installing keepalived and haproxy, you need to test whether keepalived is normal

```
root@prd-master01:~# telnet 10.0.200.220 16443
Trying 10.0.200.220...
Connected to 10.0.200.220.
Escape character is '^]'.
```

If '^]' does not appear in telnet, it is considered that VIP is unavailable and cannot be executed further down. It is necessary to troubleshoot issues with keepalived, such as the status of firewall and selinux, haproxy and keepalived, listening ports, etc.

Northward meteor forwarding N ginx

The public network mapping can be found in the following table:

<<nat_imexport_1692261237724.xlsx>>

Traffic access on the public network is accessed through Firewall's SNAT, configured with LB outside the cluster, and using two nginx for load traffic forwarding.

```
[root@k8s-ng2 conf]# cat nginx.conf
```

```
worker_processes 1;
```

```
events {
    worker_connections 1024;
}
```

```
http {
    include mime.types;

    #The core module instruction is set to binary stream by default, which is used when the file type is undefined

    default_type application/octet-stream;

    #sendfile The instruction specifies whether nginx calls the sendfile function (zero copy method) to output the file. For ordinary applications, it must be set
    to on

    #If used for heavy-load disk IO applications such as downloads, it can be set to off to balance the disk and network I/O processing speeds and reduce the
    system's uptime.

    sendfile on;

    keepalive_timeout 65; #Client long connection saves the timeout of the activity in seconds

    server {
```

```
server_name localhost;

location / {

    root html;

    index index.html index.htm;

}

error_page 500 502 503 504 /50x.html;

location = /50x.html {

    root html;

}

}

server {

    listen 8081 ssl; //Listen on port 8081, i.e. ssl

    server_name localhost;

    ssl_certificate/data/cert/8631154__holitech.net.pem; //Specify the pem file path

    ssl_certificate_key/data/cert/8631154__holitech.net.key; //Specify the key file path

    ssl_session_cache shared:SSL:1m; //Specify the session cache size

    ssl_session_timeout 5m; //Specify the session timeout

    ssl_ciphers HIGH:! aNULL:! MD5; //Specify ssl algorithm

    ssl_prefer_server_ciphers on; //Prefer server algorithm

    client_max_body_size 500M;

    #Enable reverse proxy for "/"

    location / {

        proxy_pass https://srmprd.holitech.net:30341/ ; #httpProxy, use dns background polling

        proxy_http_version 1.1; //Indicates that the version of the HTTP protocol sent by the reverse proxy is 1.1. HTTP1.1 supports long connections

    }

    #Configure support websocket start

    proxy_set_header Upgrade $http_upgrade;

    proxy_set_header Connection "Upgrade";

    #Configuration supports end of websocket

    }

}

server {

    listen 8082 ssl;

    server_name localhost;

    ssl_certificate/data/cert/8631154__holitech.net.pem;

    ssl_certificate_key/data/cert/8631154__holitech.net.key;

    ssl_session_cache shared:SSL:1m;

    ssl_session_timeout 5m;

    ssl_ciphers HIGH:! aNULL:! MD5;

    ssl_prefer_server_ciphers on;

    client_max_body_size 500M;

    location / {
```

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```
proxy_http_version 1.1;
```

```
proxy_set_header Upgrade $http_upgrade;
```

```
proxy_set_header Connection "Upgrade";
```

```
}
```

```
}
```

```
}
```

```
[root@k8s-ng2 conf]# cat/etc/keepalived/keepalived.conf
```

```
! Configuration File for keepalived
```

```
global_defs {
```

```
    router_id lb224
```

```
}
```

```
vrrp_instance paas {
```

```
    state MASTER
```

```
    interface eth0
```

```
    virtual_router_id 88
```

```
    priority 100
```

```
    advert_int 1
```

```
    authentication {
```

```
        auth_type PASS
```

```
        auth_pass 1111
```

```
    }
```

```
    virtual_ipaddress {
```

```
        10.0.200.226
```

```
    }
```

```
}
```

```
[root@k8s-ng1 ~]# cat/etc/keepalived/keepalived.conf
```

```
! Configuration File for keepalived
```

```
global_defs {
```

```
    router_id lb223
```

```
}
```

```
vrrp_instance paas {
```

```
    state BACKUP
```

```
    interface eth0
```

```
    virtual_router_id 88
```

```
    priority 50
```

```
    advert_int 1
```

```
    authentication {
```

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auth_pass 1111

in

}

virtual_ipaddress {

10.0.200.226

}

}

[root@k8s-ng1 ~]# cat/usr/local/nginx/conf/nginx.conf

worker_processes 1;

events {

worker_connections 1024;

}

http {

include mime.types;

default_type application/octet-stream;

sendfile on;

keepalive_timeout 65;

server {

listen 80;

server_name localhost;

location / {

root html;

index index.html index.htm;

}

error_page 500 502 503 504 /50x.html;

location = /50x.html {

root html;

}

}

server {

listen 8081 ssl;

server_name localhost;

ssl_certificate/data/cert/8631154__holitech.net.pem;

ssl_certificate_key/data/cert/8631154__holitech.net.key;

ssl_session_cache shared:SSL:1m;

ssl_session_timeout 5m;

ssl_ciphers HIGH:! aNULL:! MD5;

ssl_prefer_server_ciphers on;

client_max_body_size 500M;

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proxy_pass <https://srmprd.holitech.net:30341> ; #

in

proxy_http_version 1.1;

proxy_set_header Upgrade \$http_upgrade;

proxy_set_header Connection "Upgrade";

}

}

server {

listen 8082 ssl;

server_name localhost;

ssl_certificate/data/cert/8631154__holitech.net.pem;

ssl_certificate_key/data/cert/8631154__holitech.net.key;

ssl_session_cache shared:SSL:1m;

ssl_session_timeout 5m;

ssl_ciphers HIGH:! aNULL:! MD5;

ssl_prefer_server_ciphers on;

client_max_body_size 500M;

location / {

proxy_pass <http://10.0.200.210:31437> ; #

}

}

}

```
root@host-lb01:~# cat/etc/keepalived/keepalived.conf
! Configuration File for keepalived
global_defs {
    router_id LVS_DEVEL
}
vrrp_script chk_apiserver {
    script "/etc/keepalived/check_apiserver.sh"
    interval 5
    weight -5
    fall 2
rise 1
}
vrrp_instance VI_1 {
    state MASTER
    interface ens3
    # mcast_src_ip 10.0.200.206 #Send the address of the multicast package. If the default is not set, use the primary IP of the bound network card
    virtual_router_id 150 #VRIDtag, route ID, viewable via #tcpdump vrrp
    priority 101 #Priority, high priority election as master
    advert_int 2
    authentication {
        auth_type PASS
        auth_pass K8SHA_KA_AUTH
    }
    virtual_ipaddress {
        10.0.200.220/24 #Set vip
    }
    track_script {
        chk_apiserver
    }
}
```

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```
root@host-lb01:~# cat/etc/haproxy/haproxy.cfg
global
    maxconn 2000
    ulimit-n 16384
    log 127.0.0.1 local0 err
    stats timeout 30s
defaults
    log global
    mode http
    option httplog
    timeout connect 5000
    timeout client 50000
    timeout server 50000
    timeout http-request 15s
    timeout http-keep-alive 15s
frontend monitor-in
    bind *:33305
    mode http
    option httplog
    monitor-uri /monitor
frontend ks-console
    bind 0.0.0.0:80
    bind 127.0.0.1:80
    mode http
    option httplog
    default_backend qyks-console
frontend k8s-master
    bind 0.0.0.0:16443
    bind 127.0.0.1:16443
    mode tcp
    option tcplog
    tcp-request inspect-delay 5s
    default_backend qyk8s-master
backend qyk8s-master
    mode tcp
    option tcplog
    option tcp-check
    balance roundrobin
    default-server inter 10s downinter 5s rise 2 fall 2 slowstart 60s maxconn 250 maxqueue 256 weight 100
    server host-master01 10.0.200.201:6443 check
    server host-master02 10.0.200.202:6443 check
    server host-master03 10.0.200.203:6443 check
backend qyks-console
    mode http
    option httplog
    option tcp-check
    balance roundrobin
    default-server inter 10s downinter 5s rise 2 fall 2 slowstart 60s maxconn 250 maxqueue 256 weight 100
    server host-master01 10.0.200.201:30880 check
    server host-master02 10.0.200.202:30880 check
    server host-master03 10.0.200.203:30880 check
```

Host-lb01's Haproxy.cfg

```
ubuntu@host-lb02:~$ cat/etc/keepalived/keepalived.conf
! Configuration File for keepalived
global_defs {
    router_id LVS_DEVEL
}
vrrp_script chk_apiserver {
    script "/etc/keepalived/check_apiserver.sh"
    interval 5
    weight -5
    fall 2
rise 1
}
vrrp_instance VI_1 {
    state BACKUP
    interface ens3
```

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```
priority 100
advert_int 2
authentication {
    auth_type PASS
    auth_pass K8SHA_KA_AUTH
}
virtual_ipaddress {
    10.0.200.220/24
}
track_script {
    chk_apiserver
}
}
```

host-lb02's Keepalived.conf

```
ubuntu@host-lb02:~$ cat/etc/haproxy/haproxy.cfg
global
    maxconn 2000
    ulimit-n 16384
    log 127.0.0.1 local0 err
    stats timeout 30s
defaults
    log global
    mode http
    option httplog
    timeout connect 5000
    timeout client 50000
    timeout server 50000
    timeout http-request 15s
    timeout http-keep-alive 15s
frontend monitor-in
    bind *:33305
    mode http
    option httplog
    monitor-uri /monitor
frontend ks-console
    bind 0.0.0.0:80
    bind 127.0.0.1:80
    mode http
    option httplog
    default_backend qyks-console
frontend k8s-master
    bind 0.0.0.0:16443
    bind 127.0.0.1:16443
    mode tcp
    option tcplog
    tcp-request inspect-delay 5s
    default_backend qyk8s-master
backend qyk8s-master
    mode tcp
    option tcplog
    option tcp-check
    balance roundrobin
    default-server inter 10s downinter 5s rise 2 fall 2 slowstart 60s maxconn 250 maxqueue 256 weight 100
    server host-master01 10.0.200.201:6443 check
    server host-master02 10.0.200.202:6443 check
    server host-master03 10.0.200.203:6443 check
backend qyks-console
    mode http
    option httplog
    option tcp-check
    balance roundrobin
    default-server inter 10s downinter 5s rise 2 fall 2 slowstart 60s maxconn 250 maxqueue 256 weight 100
    server host-master01 10.0.200.201:30880 check
    server host-master02 10.0.200.202:30880 check
    server host-master03 10.0.200.203:30880 check
```

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